AirBnB EDA project

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Link to code:
_PClass1 AirBnB EDA.ipynb

Data Preparation

Data were processed the following way:
First, the LA and NYC datasets (39 columns and 45886 / 36111 rows, respectively) were loaded, as well as supplementary geographic datasets of NYC's borough boundaries (nynta2020.shp) and LA's Service Planning Areas (Service_Planning_Areas_2022.geojson). Each entry of the AirBnB datasets was assigned its respective area based on its geographic coordinations, in a new column named 'area'. An additional 'city' column was added to both datasets, which were then concatenated into one dataframe. Then, fifty duplicate rows were identified and removed.
Certain columns needed to be converted to appropriate data types: last_scraped, host_since, first_review and last_review were converted to datetime objects; host_is_superhost, host_has_profile_pic and instant_bookable were converted to boolean values; host_response_rate, host_acceptance_rate and price were converted to numeric values; and host_verifications and amenities were parsed into actual list objects.

Missing Data and Outliers Handling:
Rows missing the target variable, review_scores_rating, were immediately removed from the analysis (28.85% of entries). In addition, a cluster of 194 rows systematically lacked the same host properties (host_name, host_since, host_verifications, host_total_listings_count, host_listings_count and host_has_profile_pic), and were consequently removed. Missing values of other variables and extreme outliers were imputed as follow:

- Bathrooms: Missing values were assigned the mode of the bathrooms grouped by bathrooms_text. Any remaining rows that could not be imputed were removed.
- Beds & Bedrooms: data were grouped by 'accommodates', and missing values were imputed using the median value of the group.
- Host Rates: Missing host_response_rate and host_acceptance_rate were imputed with the median value (due to skewed distributions) and flagged with a new binary column (e.g., host_response_rate_imputed).
- Night Limits: Missing minimum_minimum_nights and similar columns were filled using the base minimum_nightsor maximum_nights values.
- Superhost Status: Missing values for host_is_superhost were imputed as False, under the assumption that missing data indicates the lack of Superhost status. A flagging column host_is_superhost_imputed was created to track these changes.
- Response Time: The categorical host_response_time was mapped to an ordinal scale (1-4), with missing values encoded as 0.
- Prices: 27.44% of the valid entries did not include a price. Since the price distribution was highly right-skewed, the analysis was done using the log-transformed prices, hence entries missing a price were assigned the mean log-transformed price of comparable listings (based on area and number of people the listing accommodated). They were also flagged as imputed in another dedicated column.
- Host listings: the host_total_listings_count and host_listings_count distributions were both extremely right-skewed. In a similar manner to the price column, the effect of outliers was mitigated by applying a logarithmic transformation to both columns.

Feature Engineering

In addition to the data imputations described above, a few features derived from the original features were added to the dataset:

- Has host about & has neighborhood overview: since more than 40% of entries lacked neighborhood_overview or host_about, we created these new binary variables indicating whether the original feature …..
- Property_type_grouped: This feature separated listings into 4 categories: private room, shared room, entire home/apt and 'unique', indicating a non-conventional listing.
- Host_response_time_coded - the verbal categories of host_response_time were coded numerically this way: 'within an hour': 1, 'within a few hours': 2, 'within a day': 3, 'a few days or more': 4. Entries missing host_response_time were coded as 0.
- Area: based on geographic datasets, we assigned each listing its borough/area, according to the provided coordinates.
- Description length chars/words: self-explenatory.

Hypotheses

1. Experience: More experienced hosts tend to have slightly higher ratings.

2. Demand: Higher estimated occupancy is associated with slightly higher ratings.

3. Superhost: Listings with superhost tend to have better ratings.

4. Number of listings: Hosts with more listings tend to have slightly lower ratings.

5. Entire homes and unique properties will receive higher ratings

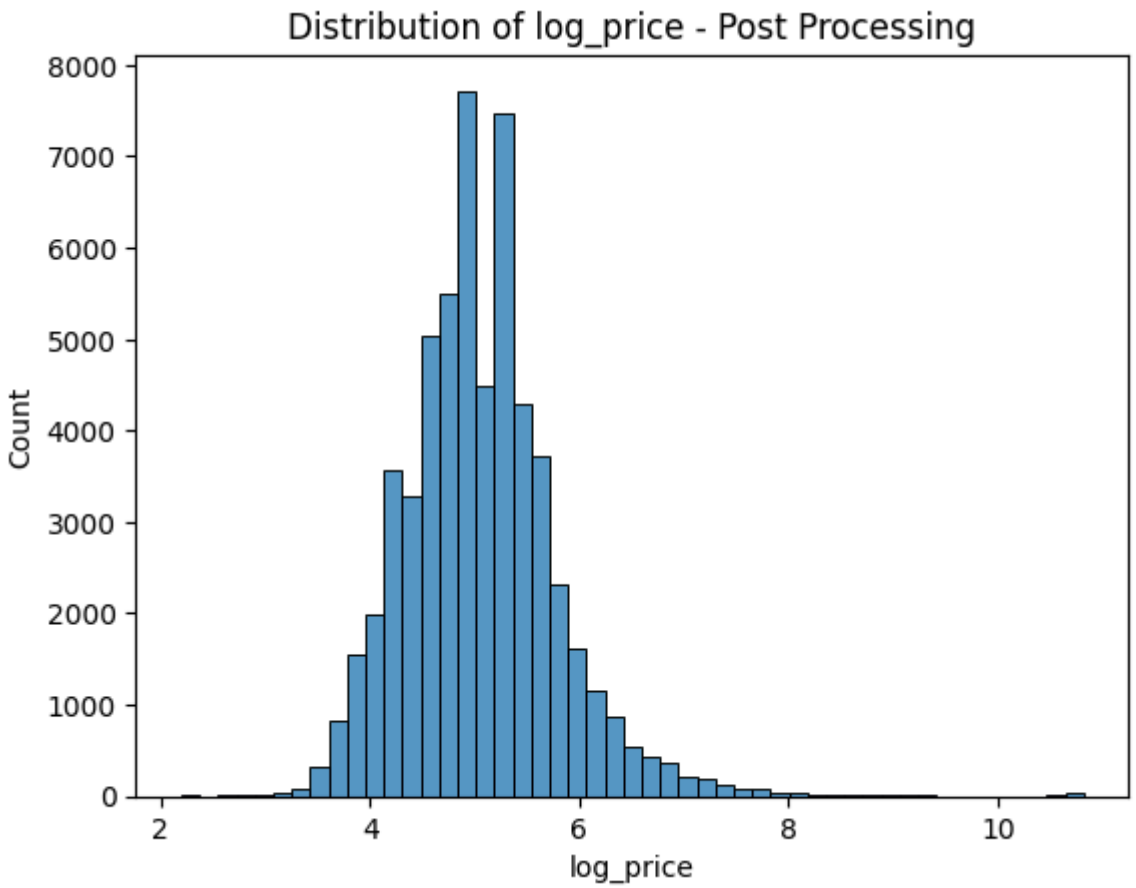
6. Host profile completeness will increase ratings on the properties

7. Adding additional bathrooms and bedrooms will increase ratings

Descriptive Statistics

- Listings and hosts are highly heterogeneous. Most hosts manage very few properties (median 2–4 listings), but a small number of professional hosts own thousands, leading to extremely high variance in host listing counts and strong right-skewness leading to our decision to do a log transformation on the data.
- Properties are generally small and moderately priced. The typical listing accommodates 2–4 guests, with around 1 bedroom and 1 bathroom. Prices are right-skewed, with a median of $150 but extreme high-price outliers, so they were processed with log-transformation (see histogram 1).
- Availability rules vary widely and contain extreme values. While median minimum and maximum stay requirements cluster around extremes of 30 (1 month) and 365 nights (1 year).

- Review scores are consistently high with limited variation. Ratings are highly concentrated near the upper bound (median 4.89/5), suggesting a ceiling effect that may limit linear predictive power.
- Host behaviour signals are strong and stable. Most hosts exhibit very high response and acceptance rates (medians near 100%), implying limited dispersion but potential usefulness as binary or threshold-based predictors.

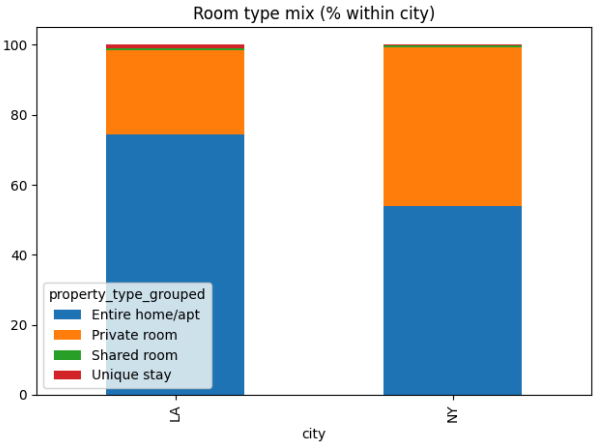


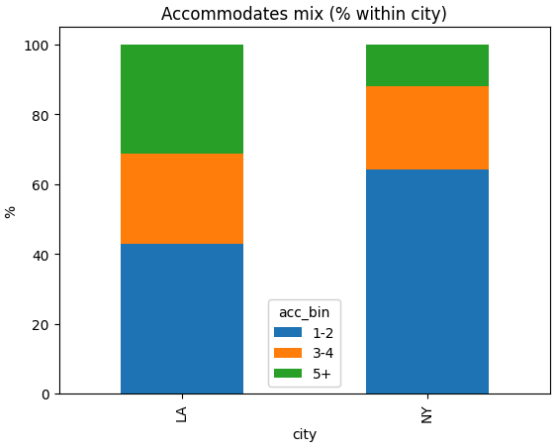
Histogram 1.

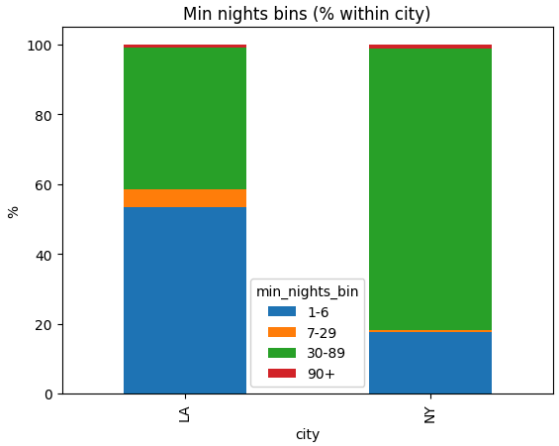
City Comparison

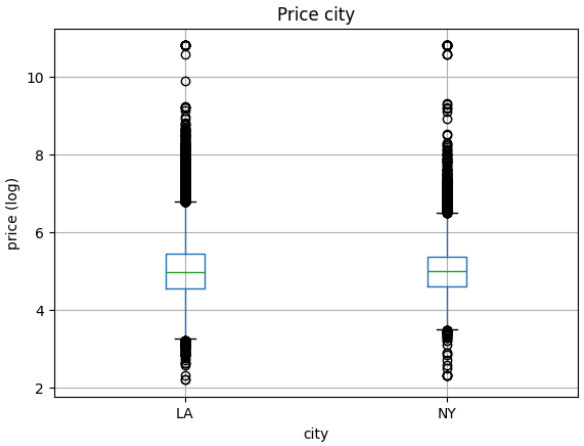
LA and NY appear to serve different guest segments. In NY, the supply is dominated by city apartments (entire units) with a relatively large share of private rooms, and many listings are configured for long minimum stays (30+ days). This pattern is consistent with smaller parties (singles/couples) and longer-term urban stays (e.g., work, study, temporary relocation). In LA, listings tend to accommodate more guests (median accommodates ≈ 3 vs ≈ 2 in NY) and short stays (1-6 days) are more common, which aligns with leisure travel and small groups/families.

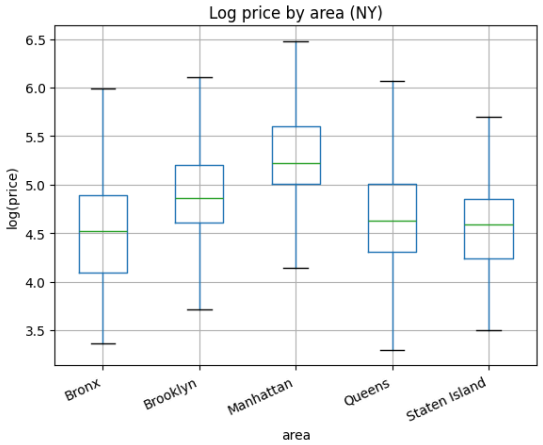
Within each city, prices vary substantially by area, confirming that location is a major driver of price. In NYC, Manhattan has the highest median log price, followed by Brooklyn and Queens, with Bronx and Staten Island lower. In LA, price differences across areas are also pronounced; however, the "Outside LA County" category shows unusually high prices and should be interpreted cautiously due to very small sample size and potential outliers.
LA has a substantially higher Superhost share than NY (~47% vs ~26%). Despite this difference in prevalence, the Superhost rating uplift is nearly identical across cities (~+0.19 points; interaction ~ 0.006), suggesting Superhost status is associated with higher ratings similarly in both LA and NY, but is more common in LA.

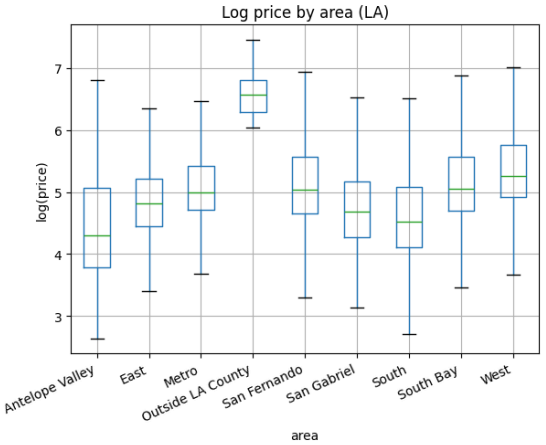


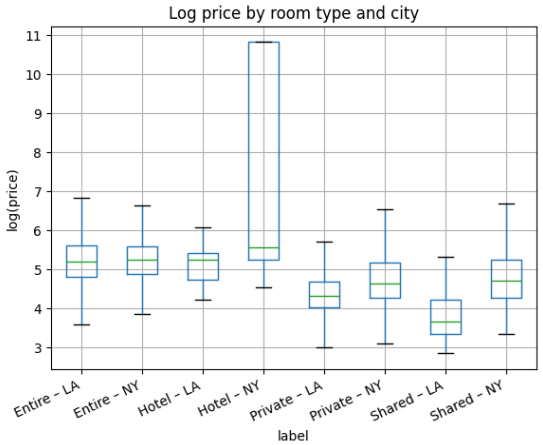


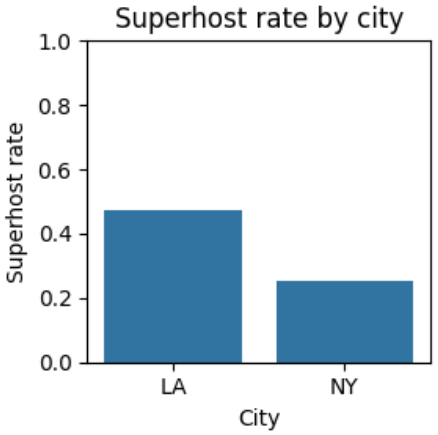





Log price by area (NY)
log(price)
area
Bronx
Brooklyn
Manhattan
Queens
Staten Island


Log price by area (LA)
log(price)
area
Antelope Valley
East
Metro
Outside LA County
San Fernando
San Gabriel
South
South Bay
West


Log price by room type and city
log(price)
label
Entire - LA
Entire - NY
Hotel - LA
Hotel - NY
Private - LA
Private - NY
Shared - LA
Shared - NY


Superhost rate by city
Superhost rate
City
LA
NY

Correlation Analysis

- Host experience (time as host, occupancy) has a weak positive association with review scores, while managing more listings slightly reduces ratings.
- Hosts with a profile picture or an “About” section, and superhosts, tend to receive higher ratings.
- Entire homes and unique properties (e.g., castles, huts, trains) generally receive higher review scores than private or shared rooms.
- More bathrooms strongly increase ratings, while additional bedrooms or beds have a modest positive effect.

Associations between numeric parameters and review score ratings

Fig. 1.

Figure 1 illustrates a correlation matrix to compare the association between review scores for different numeric parameters across Los Angeles (LA) and New York (NY) markets, and across the combined datasets. Overall, the correlations are weak, and remain so when split across the LA and NY markets. The variables that were associated with an increased review score were estimated_occupancy_l365d (Pearson’s correlation: r = 0.11; LA r = 0.12; NY r = 0.08) and time_as_host (r = 0.10; LA r = 0.11; NY r = 0.12). This indicates that being a host for longer, and having an AirBnB property that is occupied for more days of the year has a

weak, positive association with having higher review scores. In contrast, log_host_listings_count and log_host_total_listings_count were weakly negatively associated with higher review scores (Pearson's correlation: r = -0.17; LA r = -0.18; NY r = -0.19, and r = -0.18; LA r = -0.18; NY r = -0.19). This indicates that hosts with more properties are slightly associated with having lower review scores. Relationships between the aforementioned variables are also noted in figure 2.

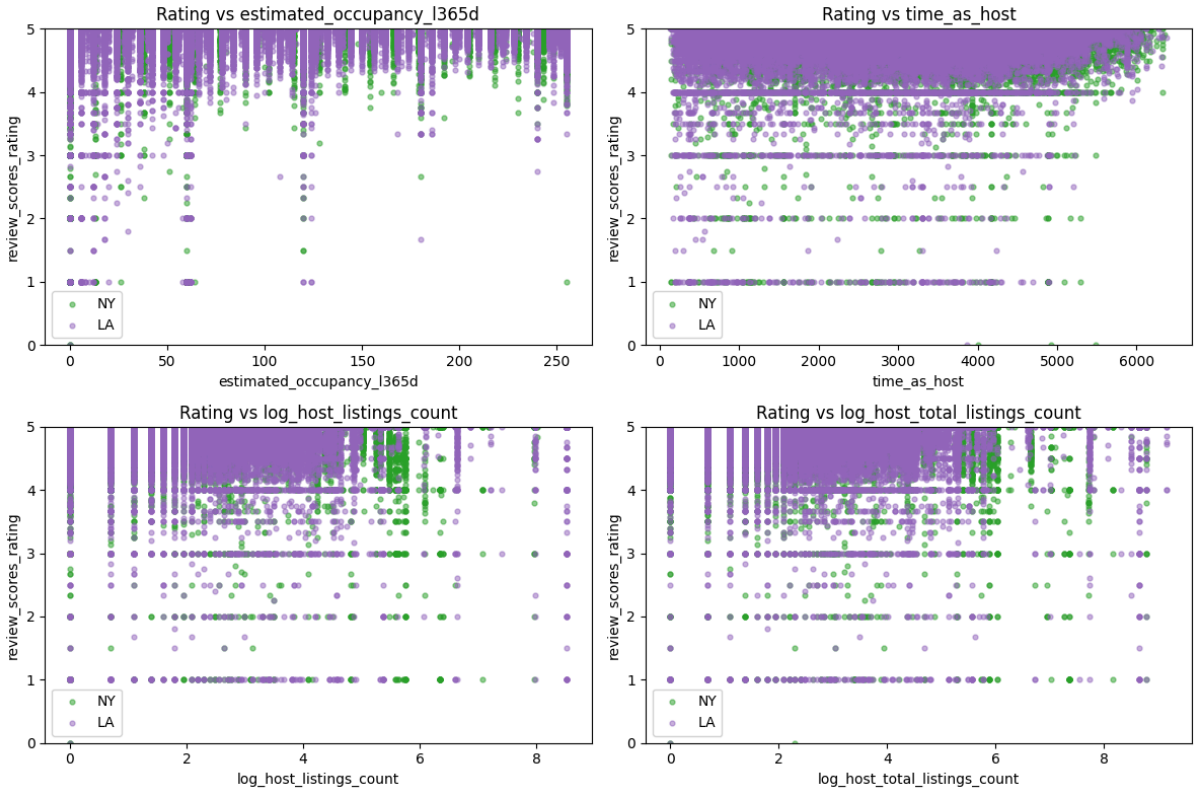


Fig. 2

Host attributes

The host attributes with the strongest correlations to review scores ratings were time being an AirBnB host (time_as_host) and the number of listings a host holds (log_host_total_listings_count, log_host_listings_count), which is detailed above. Other host attributes variables were extremely weak.

Further to this, t-tests illustrated that the differences between binary attributes such as having a profile picture (host_has_profile_pic), and including a host 'about' section (has_host_about) in the host profile were statistically significantly associated with a higher review rating (both $p < .001$).
As a sanity check, we also confirmed that superhosts, a group that AirBnB have deemed to score exceptionally well as a host, score higher on ratings ($p < 0.001$). See figure 3.

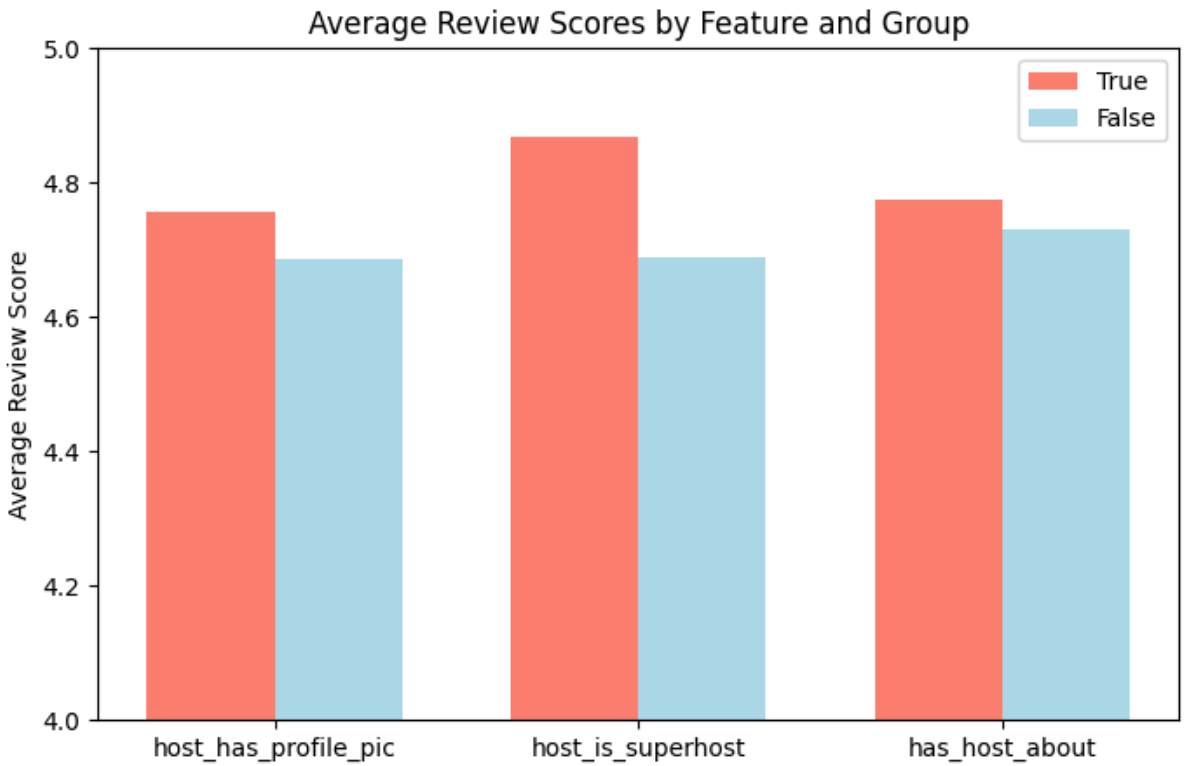


Fig. 3

Property Attributes

We examined the relationship between room type (room_type) and review scores using an ANOVA with Tukey's multiple comparisons of means. The results indicated that entire homes generally received higher ratings than other room categories. Hotel rooms also performed better than private or shared rooms, while there was no significant difference between private and shared rooms.

Next, we created a new variable (property_type_grouped) to better categorise AirBnB properties. This included a new category of 'unique' properties like castles, trains, and shepherd's huts, which might serve as a proxy for more novel or interesting experiences. In this ANOVA, 'Unique' stays received higher ratings than shared or private rooms ($p < 0.001$). Entire homes continued to receive higher ratings than private or shared rooms ($p < 0.001$). However, the differences between entire homes and Unique stays, and between private and shared rooms, were not statistically significant, suggesting that both property novelty and size can influence guest ratings. The distribution of property_type_grouped can be found later on in the report.

Another property attribute we investigated with ANOVA and Tukey comparisons was the number of bedrooms and bathrooms. We found that listings with more bathrooms tended to have higher review scores than listings with no bathrooms. This was especially true between 0 bathrooms and listings with 1-7 bathrooms ($p>0.05$). This was equal up to a < 0.32 magnitude of difference on points on the review scale by adding another, or even half a bathroom (presumably an 'en-suite'). Listings with more bedrooms also tended to have higher review scores, but the effect is weaker than for bathrooms. Significant differences are

mainly observed when comparing 0–1 bedrooms to 2–4 bedrooms, and 1–2, 1–3, 1–4 bedrooms (all $p < 0.05$), although the magnitude of difference was smaller (0.03-0.06 points). A similar, modest effect was observed by the effect of adding more beds, with some significant differences observed among 1–2, 1–3, 1–4, and 1–5 bed listings, with a magnitude of ~0.10 points.

Listing attributes

We found that properties with the ‘instant bookability’ setting had significantly lower ratings than properties that weren’t instant bookable ($p <0.001$). There was no association between actual minimum or maximum number of nights stayed and review ratings on our pearson correlation analysis.